

AIoT-based Banana Ripeness Classification Using Tiny Machine Learning for Real-Time Assessment

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Abstract- *Banana ripeness classification plays a critical role in the agro-food supply chain, as it directly affects product quality, shelf life, market acceptance, and economic losses associated with premature spoilage. Traditional ripeness assessment is commonly performed through manual visual inspection, which introduces subjectivity, inconsistencies, and dependency on human expertise. In recent years, advances in computer vision and embedded artificial intelligence have enabled the development of automated solutions for agricultural monitoring. However, many existing approaches rely on high-performance computing infrastructure, limiting their applicability in low-cost and portable environments. This study proposes an Artificial Intelligence of Things (AIoT)-based embedded system for automatic banana ripeness classification using Tiny Machine Learning. The proposed solution is implemented on the Seed Studio ESP32-S3 Sense, which integrates a camera and wireless connectivity for image acquisition and processing. A convolutional neural network model was trained using the Edge Impulse platform, where the dataset was constructed using banana images representing three ripeness stages: green, intermediate, and ripe. The trained model was optimized for TinyML deployment and executed directly on the microcontroller to enable real-time inference. Experimental validation demonstrated that the system successfully performs real-time classification of banana ripeness using images captured by the embedded camera. The classification results were displayed locally on an LCD screen, providing immediate feedback to the user. In addition, the predictions were transmitted to the cloud through the Ubidots IoT platform, enabling remote monitoring and data visualization. The results confirm the feasibility of implementing low-cost AIoT-based embedded vision systems for agricultural quality assessment. The proposed approach contributes to the automation and digitalization of post-harvest processes and demonstrates the potential of TinyML technologies for scalable smart agriculture applications.*

Keywords- *Banana ripeness classification; Tiny Machine Learning; Artificial Intelligence of Things (AIoT); Embedded computer vision.*

I. INTRODUCTION

Fruit ripening, especially banana ripening, is a fundamental process in the agro-food supply chain, as it determines the quality of the final product and its market value. In countries like Colombia, one of the largest banana exporters, accurate ripeness classification is crucial to ensure quality and reduce economic losses associated with premature deterioration. Traditionally, banana ripeness is determined through visual inspection, which introduces significant variability due to the subjectivity of the evaluator, generating inconsistencies that affect efficiency and profitability in the supply chain [1].

The use of advanced technologies such as computer vision and machine learning has emerged as an effective solution to overcome the limitations of traditional methods. Convolutional neural networks (CNNs) and IoT (Internet of Things)-based systems offer a promising avenue to automate the ripeness classification process, eliminating human variability and improving the consistency of evaluations [2]. However, many of the current developments require expensive computing platforms or specialized infrastructure, limiting their applicability in resource-constrained environments or portable systems. This is where the integration of deep learning techniques optimized for embedded devices, such as the AIoT approach with TinyML, can make a significant difference by enabling classification solutions that are accessible and real-time.

In the global context, the classification of banana ripeness not only has implications for producers but also for the entire supply chain, spanning from retail outlets to final consumers. Manual classification is inherently limited by its subjectivity, which can lead to significant inefficiencies. A study on the banana industry in Accra revealed that more than 2 million cedis (approximately 174,000 USD) are lost weekly due to over-ripening, reducing the product's value in the market [3]. Additionally, research has shown that the shelf life of bananas can vary between 18.33 and 32.67 days depending on the ripening treatment applied, highlighting how incorrect classification can drastically reduce the product's shelf life, increasing waste and economic losses in the supply chain [2].

Numerous studies have explored the use of deep learning and computer vision for classifying fruits in their various stages of ripeness. A notable example is the work in which a CNN-based system was implemented for fruit classification and ripeness evaluation. The developed system achieved 97.86% accuracy in fruit classification and nearly 99% accuracy in ripeness evaluation, using RGB images to identify the different product states. This work is relevant as it demonstrates the potential of neural networks in automating agricultural tasks, which aligns with the goals of this project [4].

Another important related work discusses an IoT system integrated with Machine Learning (ML) techniques to evaluate ripeness and detect banana spoilage in real-time. Using gas, temperature, and humidity sensors combined with ML algorithms like CatBoost, the system predicted ripeness stages with high precision, providing a viable solution to improve the supply chain and reduce losses in the industry. This research also highlights the effectiveness of combining IoT and ML, which is fundamental to the proposal of this work [1].

Finally, in the field of neural networks for ripeness classification, a solution based on MobileNetV2 for classifying 32 types of fruits and vegetables is presented, achieving 97% accuracy in classification and 99% accuracy in ripeness evaluation. This approach, which also employs deep learning, demonstrates the capability of CNN architectures to perform automated evaluations, which is relevant to the context of this project and its focus on automating banana ripeness classification [5].

This work presents an innovative solution for the automatic classification of banana ripeness using an AIoT platform based on the ESP32-S3 Sense and a machine learning model trained with Edge Impulse. The main contribution of this research is the integration of computer vision, deep learning (TinyML), and IoT to create a low-cost, high-efficiency embedded system capable of assessing banana ripeness in real-time. Through this approach, the aim is to reduce economic losses in the banana supply chain and improve classification efficiency, while minimizing the need for external computational infrastructure.

II. THEORETICAL FOUNDATIONS

The proposed solution is based on the integration of advanced technologies such as the Internet of Things (IoT) and Deep Learning, specifically optimized for embedded devices through TinyML. These technologies enable real-time classification of banana ripeness without the need for external computational infrastructure, making the system accessible, scalable, and ideal for low-cost agricultural environments.

TinyML refers to the deployment of machine learning (ML) and deep learning (DL) models on embedded devices with limited resources, such as microcontrollers and sensors. This approach has gained popularity due to its ability to run Deep Learning models on the edge, reducing implementation costs and improving operational efficiency by eliminating

latency associated with sending data to external servers [6]. In this project, the model was developed using Edge Impulse, a platform for creating ML models optimized for edge devices.

Instead of training a custom CNN from scratch, Transfer Learning was implemented using the MobileNetV2 architecture, which is well-suited for edge devices. MobileNetV2 is a lightweight and efficient convolutional neural network that leverages pre-trained weights and fine-tunes them for the specific task of banana ripeness classification. This approach significantly reduces the amount of data and computation required for training while still achieving high accuracy in classification tasks. MobileNetV2's use of depthwise separable convolutions allows it to achieve high performance with fewer parameters, making it ideal for deployment in resource-constrained environments like the ESP32-S3 Sense [7].

Once the model was trained using Transfer Learning on Edge Impulse, it was downloaded and implemented directly onto the ESP32-S3 Sense chip. The system then underwent a real-time test phase on the embedded device to evaluate the model's performance. Upon successful validation, the test results were sent to the cloud for further analysis and continuous monitoring.

In addition to local processing, the ESP32-S3 Sense is equipped with an LCD screen that provides immediate feedback to the user. The screen displays the ripeness status of the banana in real-time (e.g., green, semi-ripe, or ripe) based on the classification made by the model. This direct visual feedback allows users to assess the quality of the banana quickly and make decisions accordingly.

The integration of IoT enables real-time data capture via smart sensors, with the ESP32-S3 Sense collecting images of bananas and processing them locally. Once the ripeness state is classified, the device autonomously updates the LCD screen with the result. Furthermore, performance data from the device is sent to the cloud for continuous analysis and monitoring, ensuring that the model remains efficient and accurate, while providing valuable data for further optimization.

III. MATERIAL AND METHODS

A. Dataset

The dataset used in this project was extracted from the BananaImageBD: A Comprehensive Image Dataset of Common Banana Varieties with Different Ripeness Stages in Bangladesh database. This dataset contains labeled images of different banana varieties at various ripeness stages. For this work, the green, semi-ripe, and ripe categories were specifically used [8].

High-definition raw images were captured with the rear camera of an iPhone 13 Pro Max smartphone. The camera specification and image capture settings were as follows: resolution: 12 MP, aperture: $f/1.5$, focal length: 26 mm (wide), sensor size: 1/1.7, pixel size on the sensor: 1.9 μm , sensitivity to light: ISO 50-100, shutter speed: 1/60 s. Bananas were

detached from their stems to ensure that the entire banana is properly captured in the frame. Each banana was placed on white paper, which helped capture the object's texture, shape, and details without distortion. Furthermore, the picture was taken under white LED light to preserve the genuine color of the bananas, ensuring that no shadows were visible. Once the banana was correctly positioned, the image was captured from a specific height to confirm that the entire banana was visible in the frame. After photographing one side of the banana, it was rotated to capture images of the other sides of the banana. This process was repeated to take pictures of each banana from all four sides, helping improve the dataset's diversity and generalization [8].

The original dataset contains 617 images, with a distribution of 212 images for the "green" category, 204 for "semi-ripe," and 201 for "ripe." To improve the model's performance and increase the variability in the training data, data augmentation techniques were applied, resulting in a total of 1848 augmented images. The final distribution is shown in Table 1.

TABLE I
Distribution of Original and Augmented Images for
Banana Ripeness Classification

Class	Number of original images	Number of data-augmented images	Percentage
Green	212	636	34%
Semi-ripe	204	609	33%
Ripe	201	603	33%
Total	617	1848	100%

B. Data Augmentation

Data augmentation plays a crucial role in introducing diversity and generalization within datasets, especially in real-world scenarios where it is impossible to capture all possible variations or perspectives of the data. Raw datasets usually contain limited examples, often showing a single version of the object or scenario. When machine learning (ML) or deep learning (DL) models are trained on such data, they are prone to overfitting, meaning the model becomes too specialized to the training data and lacks the ability to generalize to new, unseen instances of the same object under different conditions.

To address the problem of limited data diversity, augmentation methods are used to artificially create new variations of the existing data. This process allows for the expansion of the dataset without the need for extensive manual data collection. By applying data augmentation techniques, raw datasets can be enriched with different perspectives and variations of the data, enabling models to learn features from multiple viewpoints. This ultimately helps the model generalize better and become more robust in recognizing objects in diverse, complex, and real-world conditions [8].

In the case of the BananalImageBD dataset, several augmentation techniques were already applied to the images. These techniques were used to create additional variations of

the images, resulting in a larger and more diverse dataset. Each original image was augmented to produce two additional images, resulting in a ratio of augmented images to original images of 2:1. The following augmentation methods were employed:

- 1) **Flipping:** Images were flipped horizontally and vertically to simulate different orientations.
- 2) **90° Rotation:** Images were rotated in 90-degree increments (clockwise, counter-clockwise, and upside-down) to help the model recognize objects regardless of their orientation.
- 3) **Rotation:** A small range of rotations was applied (between -15° and $+15^\circ$) to create slight variations in the image while maintaining the object's integrity.
- 4) **Brightness:** The brightness of images was adjusted between -15% and $+15\%$ to account for different lighting conditions in real-world scenarios.
- 5) **Blur:** A blur effect was applied to some images, with a maximum blur of 2.5 pixels, to simulate potential distortions.
- 6) **Noise:** Random noise was added to images, with a maximum noise level of 1.5% of pixels, to help the model learn to distinguish relevant features despite visual distortions.

These augmentation methods, as specified in the original dataset, helped diversify the raw data and improve the generalization and robustness of models trained on this dataset [8].

C. Model Development and Configuration

The machine learning model was developed using MobileNetV2, a lightweight Convolutional Neural Network (CNN) architecture that is widely used for image classification tasks, especially when optimized for resource-constrained environments. MobileNetV2 is specifically designed to be efficient and suitable for on-device vision applications, making it ideal for embedded systems where computational resources are limited. The images were processed with an RGB color depth, allowing the model to capture the full chromatic information necessary to distinguish between the different ripeness stages of the bananas. These images were resized to a 96×96 pixel resolution, ensuring consistency across all inputs while optimizing processing speed without losing important details for classification.

For model optimization, a quantized int8 model was used. This optimization increases on-device performance by reducing the model's size and computational requirements, although it may slightly reduce accuracy. Performance estimates for an 8-bit quantized model running on microcontrollers like the ESP32-S3 show that such models can achieve efficient inference while fitting within the limited memory available on embedded platforms. The model's RAM usage is 334.7 K and the flash memory usage is 535.7 K,

reflecting the successful adaptation of the model to the constraints of the ESP32-S3 Sense.

The model was trained using the CPU as the processing unit, with 60 training cycles (epochs) to ensure comprehensive learning. The learning rate was set to 0.0005 to facilitate smooth convergence. A validation set size of 20% was used to ensure that the model generalizes well and avoids overfitting. The batch size was set to 32 to balance memory usage and model performance during training.

The model was configured to classify banana ripeness into three categories: green, semi-ripe, and ripe. This was achieved by defining the output features in the classification module of the training pipeline, where the model learned to differentiate the three ripeness stages based on the visual features extracted from the banana images.

IV. RESULTS AND DISCUSSION

A. Training

The neural network architecture of the model consists of an input layer with 27,648 features, corresponding to the RGB images resized to 96x96 pixels. The network uses MobileNetV2 as the backbone architecture, which is optimized for efficient performance in embedded systems. The final layer of the network has 16 neurons and a dropout rate of 0.1 to reduce overfitting and improve generalization. The model is designed to classify the ripeness of bananas into 3 categories: Green, Semi-ripe, and Ripe. The output layer, therefore, has 3 units to represent these classes.

The performance of the model was evaluated using several key metrics, including Accuracy, Precision, Recall, and F1-Score, each of which provides different insights into the model's performance in classifying banana ripeness [9].

Accuracy is one of the main evaluation metrics, which measures the proportion of correct predictions relative to the total number of observations. It is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Where TP is True Positives, TN is True Negatives, FP is False Positives and FN is False Negatives. The model achieved an Accuracy of 84.5%, indicating that the percentage mentioned of the predictions made by the model were correct, out of all the images classified.

Precision is another critical metric that assesses the proportion of positive predictions that were actually correct. It is given by:

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

In the current model, the weighted average Precision was 0.86, which suggests that the model is highly reliable in predicting the correct ripeness stages (Green, Semi-ripe, Ripe) when it classifies a banana as a certain category.

Recall measures the ability of the model to identify all relevant instances within the data. It is calculated as:

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

The weighted average Recall for the model was also 0.84, indicating that the model was effective at identifying most of the instances for each class, minimizing the number of False Negatives.

F1-Score is a metric that combines both Precision and Recall, providing a balance between the two. The F1-Score is calculated as:

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

The model achieved an F1-Score of 0.85, demonstrating a strong balance between Precision and Recall, making it a reliable classifier overall.

On the other hand, The Confusion Matrix provides a detailed breakdown of the model's classification performance across all classes (Fig. 1.). For the validation set, the matrix shows the following performance: For the Ripe class, the model correctly classified 82.7% of the instances, with only 17.3% misclassified as Semi-ripe. For the Semi-ripe class, the model correctly identified 84.8%, with 12.1% misclassified as Rip and 3% as Green. For the Green class, the model achieved 85.9% accuracy in classifying this category, with 14.1% misclassification as Semi-ripe.

	GREEN	RIPE	SEMI-RIPE
GREEN	85.9%	0%	14.1%
RIPE	0%	82.7%	17.3%
SEMI-RIPE	3.0%	12.1%	84.8%
F1 SCORE	0.91	0.85	0.79

Fig. 1. Confusion Matrix for the Training Set in Banana Ripeness Classification

This confusion matrix highlights the model's ability to correctly classify the Green class with high precision, while the Ripe class has a slightly lower precision due to some misclassifications such as Semi-ripe.

The Data Explorer plot (Fig. 2.) further illustrates the correct and incorrect classifications in the full training set. The scatter plot visually demonstrates the separation between the classes, with green circles representing correctly classified instances and red circles representing incorrect classifications. The model's decision boundary appears to be well-formed, as most of the correct instances are clustered, although there are some misclassified instances in the overlap areas.

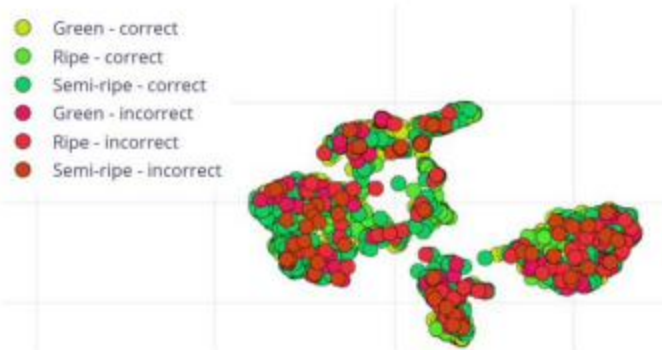


Fig. 2. Data Explorer Visualization for the Training Set in Banana Ripeness Classification

B. Testing

After training, the model was evaluated on a separate test set, and several metrics were calculated to assess its performance in real-world conditions. The model achieved an Accuracy of 89.19%, meaning a significant majority of the predictions were correct. Accuracy is an important metric as it provides an overall measure of how well the model performs across all classes.

In addition to accuracy, other metrics such as Precision, Recall, and F1-Score were also considered to comprehensively assess the model's performance. The weighted average Precision was 0.92, the Recall was 0.92, and the F1-Score was 0.92. These values indicate that the model was effective in making reliable predictions across the three ripeness categories, with a balanced performance in terms of both precision and recall.

The Confusion Matrix (Fig. 3.) provided a more detailed breakdown of the classification results. For the Green class, the model correctly classified 96% of the instances as Green, with very few misclassifications as Semi-ripe or Uncertain. For the Ripe class, the model correctly identified 88.6% of the instances, with a small portion misclassified as Semi-ripe. The Semi-ripe class had a 82.8% of instances classified correctly, and misclassification as Ripe and Uncertain.

	GREEN	RIPE	SEMI-RIPE	UNCERTAIN
GREEN	96%	0%	3.2%	0.8%
RIPE	0%	88.6%	8.1%	3.3%
SEMI-RIPE	3.3%	4.9%	82.8%	9.0%
F1 SCORE	0.96	0.92	0.85	

Fig. 3. Confusion Matrix for the Test Set in Banana Ripeness Classification

Lastly, the Data Explorer plot (Fig. 4.) visually represents the model's performance on the test set. The plot shows correctly classified instances marked in green and misclassified instances marked in red. It reveals that the majority of correct classifications are clustered, although there are some misclassifications, particularly in the overlap areas between Ripe and Semi-ripe categories.

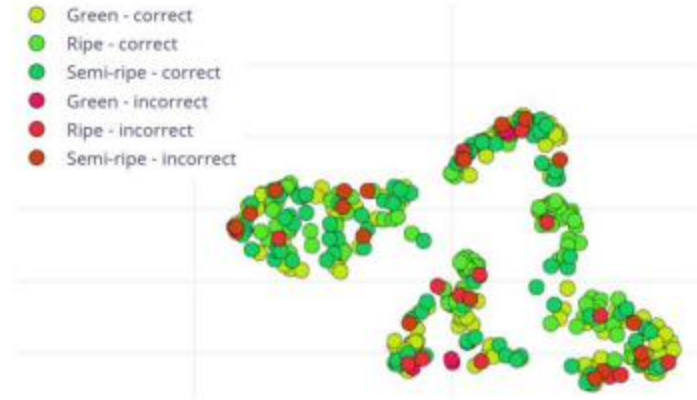


Fig. 4. Data Explorer Visualization for the Test Set in Banana Ripeness Classification

C. Discussion

The model's performance in this study provided valuable insights into its effectiveness for banana ripeness classification. During the training phase, the model achieved an accuracy of 84.5% on the validation set, with Precision and Recall values of 0.86 and 0.84, respectively. These values reflect the model's ability to accurately classify banana ripeness into three categories: Green, Semi-ripe, and Ripe, while minimizing errors. The F1-Score during training was 0.85, indicating a well-balanced performance in terms of precision and recall.

In the confusion matrix for the training set, the model demonstrated its strongest performance in classifying Green bananas, with an impressive 85.9% accuracy. However, it struggled more with the Ripe category, where only 82.7% of instances were correctly classified, and Semi-ripe bananas were classified with an accuracy of 84.8%. This shows that the model is particularly effective at distinguishing Green bananas, but there is room for improvement in differentiating Semi-ripe from Ripe bananas.

The test phase results showed an accuracy of 89.19% for the model, with a weighted average Precision, Recall, and F1-Score all at 0.92. These results demonstrate a significant improvement over the training phase, which could be attributed to the use of Transfer Learning with MobileNetV2, allowing the model to leverage pre-trained weights and fine-tune them for banana ripeness classification. The confusion matrix for the test set revealed that the model performed exceptionally well in classifying Green bananas with 96% accuracy, but there was still some confusion in classifying Ripe and Semi-ripe bananas, with 88.6% accuracy for Ripe and 82.8% for Semi-ripe.

The trained model was successfully deployed on the ESP32-S3 Sense microcontroller, enabling real-time inference directly on the embedded device. During the experimental validation stage, the system demonstrated the ability to correctly classify banana ripeness into the predefined categories (green, intermediate, and ripe) using images captured by the onboard camera. The classification results were displayed immediately on the integrated LCD screen, providing direct visual feedback to the user. Additionally, the predicted ripeness state was transmitted to the cloud using the Ubidots IoT platform, allowing remote visualization and monitoring of the classification results. The tests confirmed

that the proposed AIoT system can perform reliable ripeness classification in real time while simultaneously providing both local feedback and cloud-based data reporting, demonstrating the practical applicability of the system in smart agricultural monitoring scenarios.

In comparison to other research, a project achieved 90.42% accuracy using high-resolution images captured under controlled conditions, the results here are slightly lower but still competitive. One of the key differences lies in the trade-off between model performance and deployment constraints. While the study used high-resolution images and a more complex CNN architecture, the approach of this study focuses on making the model feasible for real-world deployment in agricultural settings, where cost, portability, and autonomy are crucial. The model's ability to operate in resource-constrained environments, providing real-time feedback via the LCD screen, is a critical advantage that cannot be overlooked [10].

In a similar vein, other studies reported higher accuracy rates (close to 97%) [11]. However, these studies typically rely on large datasets and the use of data augmentation techniques, which may not always be feasible for small-scale applications or environments with limited resources. While the performance in this study may not surpass those figures, the focus on embedded, low-cost solutions represents a significant advancement in terms of practical implementation and scalability. The system's ability to provide immediate, on-site classification makes it highly applicable for use by farmers, where decisions on harvesting and quality control can be made without waiting for cloud processing or external infrastructure.

V. CONCLUSIONS

This work presented the design and implementation of an AIoT-based embedded system for automatic banana ripeness classification using Tiny Machine Learning. The proposed solution integrates computer vision and embedded artificial intelligence to perform real-time classification of banana ripeness directly on a low-cost microcontroller.

The system uses the Seeed Studio ESP32-S3 Sense platform, which integrates an onboard camera and wireless capabilities, enabling image acquisition and local processing without relying on external computing infrastructure. A convolutional neural network model was trained using the Edge Impulse platform and optimized for TinyML deployment, allowing the model to run efficiently within the hardware constraints of the microcontroller.

Experimental validation demonstrated that the system is capable of successfully classifying banana ripeness into three categories: green, intermediate, and ripe. The trained model was deployed on the ESP32-S3 Sense and tested with new samples, confirming that real-time inference is feasible in an embedded environment. The classification result was displayed immediately through an LCD interface, enabling direct interaction with the user.

The results confirm the technical feasibility of implementing computer vision-based agricultural monitoring systems using low-cost embedded hardware. This approach reduces dependence on high-performance computing infrastructure and enables portable solutions suitable for real-world agricultural environments.

Future work will focus on expanding the dataset with a larger number of banana samples captured under more diverse environmental conditions, including variations in illumination, background complexity, and fruit positioning, in order to improve the robustness and generalization capability of the model. Additionally, further research could explore the use of more advanced lightweight neural network architectures and model optimization techniques to enhance classification accuracy while maintaining efficient execution on embedded hardware. Another potential direction involves evaluating the system in real post-harvest environments such as packing facilities or distribution centers to assess its performance under operational conditions. Finally, the proposed approach could be extended to the classification of ripeness in other fruits, contributing to the development of scalable AIoT-based solutions for automated quality assessment in agricultural supply chains.

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